

SIH 26048

iKwath

Expert-Level Winning Solution Blueprint

Pod-based smart Kwatha maker

A formulation-aware, gravimetrically controlled, accelerated mild-temperature decoction platform.

Category: Hardware Theme: MedTech / BioTech / HealthTech

Organization: Ministry of Ayush Department: All India Institute of Ayurveda

Core winning idea

Do not pitch this as an automatic Kadha maker. Pitch it as an engineered process-control platform that accelerates classical preparation while validating that the accelerated output remains within a predefined reference profile.

Prepared as a practical SIH strategy document. Prototype claims and targets should be experimentally validated before being presented as facts.

1. PROBLEM DECODED

The challenge is not simply to heat herbs. The machine must automate soaking, controlled mild-temperature extraction, reduction from roughly 400 mL toward a single roughly 100 mL dose, filtration and dispensing - while maintaining dose consistency and a validated quality target.

The engineering question

How can water be removed faster without relying on excessive temperature, while preserving the intended extraction conditions and final reduction endpoint?

Core process: Dose -> Soak -> Extract -> Heat -> Reduce -> Filter -> Dispense

Hidden difficulty: A fixed timer is not enough. Evaporation changes with formulation, ambient conditions, heat transfer and surface conditions, so the machine should measure actual process state.

2. WHAT THE GOVERNMENT REALLY NEEDS

- Dose consistency: repeatable output from a standardized single-dose pod.
- Process consistency: controlled temperature, water proportion and reduction endpoint.
- Formulation flexibility: one appliance supporting multiple validated formulation profiles.
- Fresh preparation: brew on demand rather than relying on concentrated shelf-stable products.
- Easy cleaning and safety: removable food-contact parts, rinse workflow, anti-boil-over and dry-run protection.
- Evidence: demonstrate that accelerated preparation reproduces the reference process within predefined, formulation-specific criteria.

3. CURRENT APPROACH & GAPS

Current approach	Main gap	Opportunity for iKwath
Manual preparation	Measurement and reduction depend on the user	Closed-loop measurement and control
Timer-based appliance	Same timer may not produce same endpoint	Gravimetric endpoint
High-power heating	Can change thermal exposure	Increase evaporation effectiveness while controlling temperature
One fixed recipe	Poor flexibility across formulations	Pod-specific process profile
Manual filtering/cleanup	Messy and inconsistent	Integrated removable filter + spent pod

4. COMPETITOR / COMMON TEAM APPROACHES

Likely team idea	Why it is weak	How to beat it
ESP32 + heater + timer	Looks like an automated kettle	Add mass-based endpoint and validation
AI-powered Kadha maker	AI may be cosmetic	Use data only where it improves process control
IoT dashboard	Low direct value to the core problem	Make the physical process itself the innovation
Pressure-cooker style system	Changes thermal/pressure environment	Maintain validated mild-temperature process

Likely team idea	Why it is weak	How to beat it
Camera herb identification	Pod already identifies formulation	Use pod metadata for formulation-aware control

5. 5-10 POSSIBLE SOLUTIONS

Solution	Core concept	Prototype cost	Innovation /10	Feasibility /10	SIH potential /10
A. Smart PID reduction chamber	Heater + RTD + load cell; endpoint by mass	₹8k-15k	7	9	8
B. Recirculating thin-film system	Circulate decoction through high-surface-area evaporation region	₹12k-25k	9	8	9.5
C. Reduced-pressure evaporation	Vacuum-assisted low-temperature evaporation	₹20k+	10	5	8
D. High-surface-area chamber	Shallow liquid layer + controlled vapour removal	₹7k-15k	8	9	9
E. Hybrid adaptive reduction	Recirculation + thin film + mass endpoint + formulation profile	₹15k-25k	9.5	8	9.5

6. WINNING SOLUTION

Proposed product: iKwath-X - Adaptive Thin-Film Automated Kwatha Preparation System

The recommended architecture combines formulation-aware pods, controlled extraction, recirculation, increased evaporation surface, continuous mass monitoring and an integrated filter/dispense stage.

System flow

FORMULATION POD -> POD IDENTIFICATION -> WATER MANAGEMENT -> SOAKING -> CONTROLLED EXTRACTION -> RECIRCULATION -> THIN-FILM EVAPORATION -> MASS ENDPOINT -> FILTER -> DISPENSE

7. WHY THIS CAN WIN SIH

- It solves the actual engineering bottleneck: controlled reduction, not just heating.
- It has a strong physical WOW factor: a visible automated process with live mass/temperature data.
- It supports a convincing evidence story: classical/reference batch versus machine batch.
- It is scalable: the same hardware can support multiple validated formulation profiles.
- It is cost-conscious and achievable with common embedded components.
- It creates a clear path from student prototype to a regulated, formulation-specific product development program.

8. COMPLETE SYSTEM ARCHITECTURE

INPUT -> PROCESSING -> INTELLIGENCE -> ACTION -> OUTPUT

Layer	Implementation
Input	Pod ID, powder dose, water quantity, temperature profile, target endpoint
Sensing	PT100/PT1000 RTD, load cell + HX711, optional flow/level sensing
Processing	ESP32-S3 control logic and formulation profile database
Intelligence	Closed-loop temperature + mass control; optional evaporation trajectory prediction
Actuation	Heater, recirculation pump, valves, filtration/dispense mechanism
Output	Fresh warm decoction + cycle record + process status

9. HARDWARE

Component	Recommended choice	Purpose
Controller	ESP32-S3	Control, display, connectivity and data logging
Temperature	PT100/PT1000 + MAX31865	Accurate measurement in the operating range
Mass	Load cell + HX711	Track actual water removal and endpoint
Heater	Food-safe controlled heating element	Mild-temperature extraction / reduction
Pump	Food-compatible low-flow pump	Recirculation through thin-film section
Valves	Food-grade solenoid/pinch valves	Fluid routing and dispense control
Pod ID	NFC or QR	Load formulation-specific process profile
Chamber	SS304; selected parts may use 316L	Food-contact vessel
Filter	Removable food-grade mesh/filter	Retain coarse powder and improve clarity
Safety	Thermal cutoff, fuse, dry-run/level logic, lid/interlock	Prevent unsafe operation

10. SOFTWARE

Core firmware sequence

Read pod profile -> verify chamber/water state -> measure initial mass -> soak -> controlled heating -> recirculate -> monitor temperature and mass -> adapt process -> stop at calibrated endpoint -> filter -> dispense -> log cycle.

Important design choice

Do not use a simple 'if timer > 30 minutes, stop' rule. Use measured process state, especially mass, as the primary reduction endpoint.

Optional user interface: a small screen can show formulation, temperature, initial/current mass, target mass, progress and status. A mobile/cloud layer is secondary and should not distract from the physical innovation.

11. AI / ML

AI should be optional and introduced only after deterministic control works. A useful role is predictive evaporation control: learn from temperature, mass, evaporation rate, formulation and ambient conditions to estimate time-to-endpoint.

Avoid claiming that AI determines medicinal concentration. A cheap TDS sensor is not a validated measure of phytochemical concentration.

12. INNOVATIVE FEATURES

Innovation	Existing approach	Our approach	Demonstration
1. Gravimetric endpoint	Timer	Actual mass reduction	Live 400 g -> target mass graph
2. Formulation-aware pod	Fixed recipe	Pod loads validated parameters	Swap pods and show changed profile
3. Thin-film acceleration	Deep liquid layer	High effective evaporation surface + recirculation	Compare cycle time
4. Reference fingerprint	Visual inspection	Reference vs machine analytical/optical comparison	Side-by-side data
5. Removable cleaning module	Manual cleaning	Tool-free chamber/filter removal + rinse	Timed cleaning demo

13. BILL OF MATERIALS

Component	Qty	Approx. cost	Purpose
ESP32	1	₹600	Controller
PT100 + MAX31865	1	₹1,000	Temperature
Load cell + HX711	1	₹500	Mass measurement
Heating element + control	1	₹1,000	Heating
Food-compatible pump	1	₹500-1,000	Recirculation
Valves/connectors/tubing	Set	₹1,000-1,500	Fluid routing
NFC/QR reader	1	₹300-800	Pod identification
SS chamber	1	₹1,500-3,000	Food-contact vessel
Filter assembly	1	₹500-1,000	Filtration
Display	1	₹300-700	UI
Power/safety hardware	Set	₹1,000-1,500	Protection
Mechanical enclosure/misc.	Set	₹2,000-4,000	Prototype integration

Target: Minimum proof-of-concept roughly ₹9k-11k. Recommended SIH prototype roughly ₹15k-25k. Prices are planning estimates and should be verified locally before procurement.

14. PROTOTYPE DEVELOPMENT PLAN

Phase	Tasks	Expected output	Cost	Priority
Phase 1 - Reference process	Select one formulation; establish classical/reference preparation; record mass, temperature, time and yield.	Reference dataset	Low	Highest priority
Phase 2 - Automated chamber	Build chamber, heater, RTD, load cell and ESP32 control.	Repeatable controlled reduction	₹5k-10k	Core hardware
Phase 3 - Acceleration	Add recirculation + thin-film section; compare cycle time and quality metrics.	Validated acceleration candidate	₹5k-10k	Critical experiment

Phase	Tasks	Expected output	Cost	Priority
Phase 4 - SIH demonstrator	Integrate pod ID, display, safety, filter and polished enclosure.	Live product-style demo	₹5k-10k	Presentation-ready

15. TESTING & KPIs

Establish engineering targets first; do not present them as official API/AFI acceptance limits unless the applicable monograph explicitly states them.

KPI	Prototype target / method
Final mass consistency	Aim initially for +/-5%; improve through calibration
Temperature control	Aim for approximately +/-1-2°C around the selected setpoint
Dose repeatability	At least 95% of engineering target across repeated runs
Cycle-time reduction	Measure against the same reference formulation and conditions
Cleaning time	Target <3 minutes for removable parts
Manual intervention	Minimal after loading water and pod
Safety shutdown	100% pass in deliberate fault tests
Quality equivalence	Compare extractive yield plus appropriate formulation-specific analytical markers

Validation design: Prepare multiple reference batches and multiple iKwath batches. Compare final mass, extractive yield, temperature-time profile, pH, suitable rapid proxies, optical fingerprint and selected phytochemical markers where appropriate. Formal analytical validation should use an appropriate laboratory method such as HPLC/HPTLC when applicable.

16. REAL-WORLD DEPLOYMENT

- Home: one pod, one fresh dose, minimal user interaction.
- Ayurvedic clinic: multiple validated formulations and cycle records.
- Dispensary/pharmacy: pod inventory + standardized preparation workflow.
- Manufacturing: stainless food-contact path, replaceable pump/filter, modular electronics.
- Reliability: fault detection, thermal protection, calibration and preventive cleaning.
- Connectivity: appliance should remain functional offline; connectivity is an enhancement, not a dependency.
- Data security: store only required process information and secure any networked features.

17. SIH DEMONSTRATION PLAN

1. Show the pod. 2. Insert it. 3. Display the automatically loaded formulation profile. 4. Show live temperature and mass reduction. 5. Demonstrate endpoint detection. 6. Filter and dispense. 7. Show reference-vs-machine validation data.

Best visual moment

Put a live progress display beside the transparent/visible process section: FORMULATION -> TEMP -> CURRENT MASS -> TARGET MASS -> PROCESS STATUS. The judge should be able to understand the machine without reading your poster.

18. JUDGE QUESTIONS & ANSWERS

Judge question	Strong answer
Why not a pressure cooker?	It changes the thermal/pressure environment. Our design targets formulation-specific controlled extraction and reduction conditions.
Why is it faster?	We investigate increased effective evaporation surface and controlled vapour removal/recirculation rather than simply raising temperature.
How do you know it is equivalent?	We do not claim equivalence without evidence. We compare repeated accelerated batches with a reference preparation using predefined analytical criteria.
Why use a pod?	It standardizes dose, reduces handling and enables formulation-specific parameters.
Why not a timer?	Mass directly measures reduction; a timer assumes evaporation is always identical.
Does TDS prove medicinal concentration?	No. It can be a process proxy, but validated phytochemical concentration requires appropriate analytical testing.

19. RISKS & FAILURE MODES

Risk	Mitigation
Cycle is not fast enough	Run evaporation experiments before freezing mechanical design; optimize surface area and recirculation.
Different formulations behave differently	Create formulation-specific validated process profiles.
Foaming/clogging	Test representative formulations; add flow/level protection and suitable filtration.
Quality changes during acceleration	Use reference-batch experiments; reject any acceleration method that fails predefined criteria.
Sensor drift	Calibration routines and plausibility checks.
Cleaning becomes burdensome	Design removable food-contact module from the start.
Overclaiming equivalence	Clearly distinguish engineering prototype evidence from formal regulatory/analytical validation.

20. JUDGE SCORE /100

Category	Initial expert score
Problem Understanding	9/10
Innovation	18/20
Technical Feasibility	13/15
Impact	14/15
Prototype Quality	12/15
Scalability	9/10
Cost Effectiveness	5/5
Presentation	4/5
Uniqueness	5/5
TOTAL	89/100

Potential after strong experimental validation: roughly 93-96/100 potential. Without validation: the concept risks falling into the 70-80 range because the core claim remains unproven.

21. FINAL EXPERT VERDICT

Recommended SIH positioning

Do not call it an automatic Kadha maker.

Position it as: "A formulation-aware, gravimetrically controlled, accelerated mild-temperature decoction platform that reproduces a validated classical Kwatha preparation with automated reduction and dose consistency."

The pod + NFC + ESP32 is the interface. The real technical heart is controlled extraction + accelerated reduction + gravimetric endpoint + reference-batch validation.

Priority before procurement: Build a small experimental rig first. Determine whether thin-film recirculation actually reduces the 400 -> 100 mL cycle time without materially changing extractive yield or the selected quality profile. Let that experiment determine the final mechanical design.

Official reference: PCIM&H; / Ministry of Ayush material should be consulted for the applicable formulation and pharmacopoeial/formulary requirements before making any compliance claim.